# Lec-16

25/10/26 4:48 PM

## Fibo

0 1 1 2 3 5 8 13 21 34 55 89 — — —

n==0 n==1

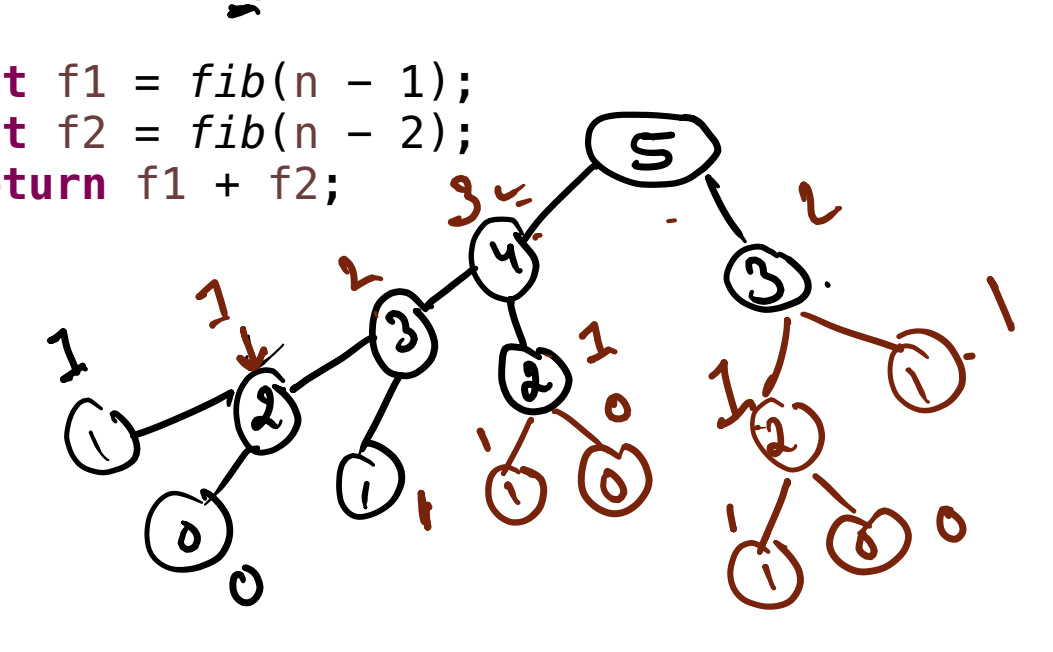
Fn = Fn-1 + Fn-2

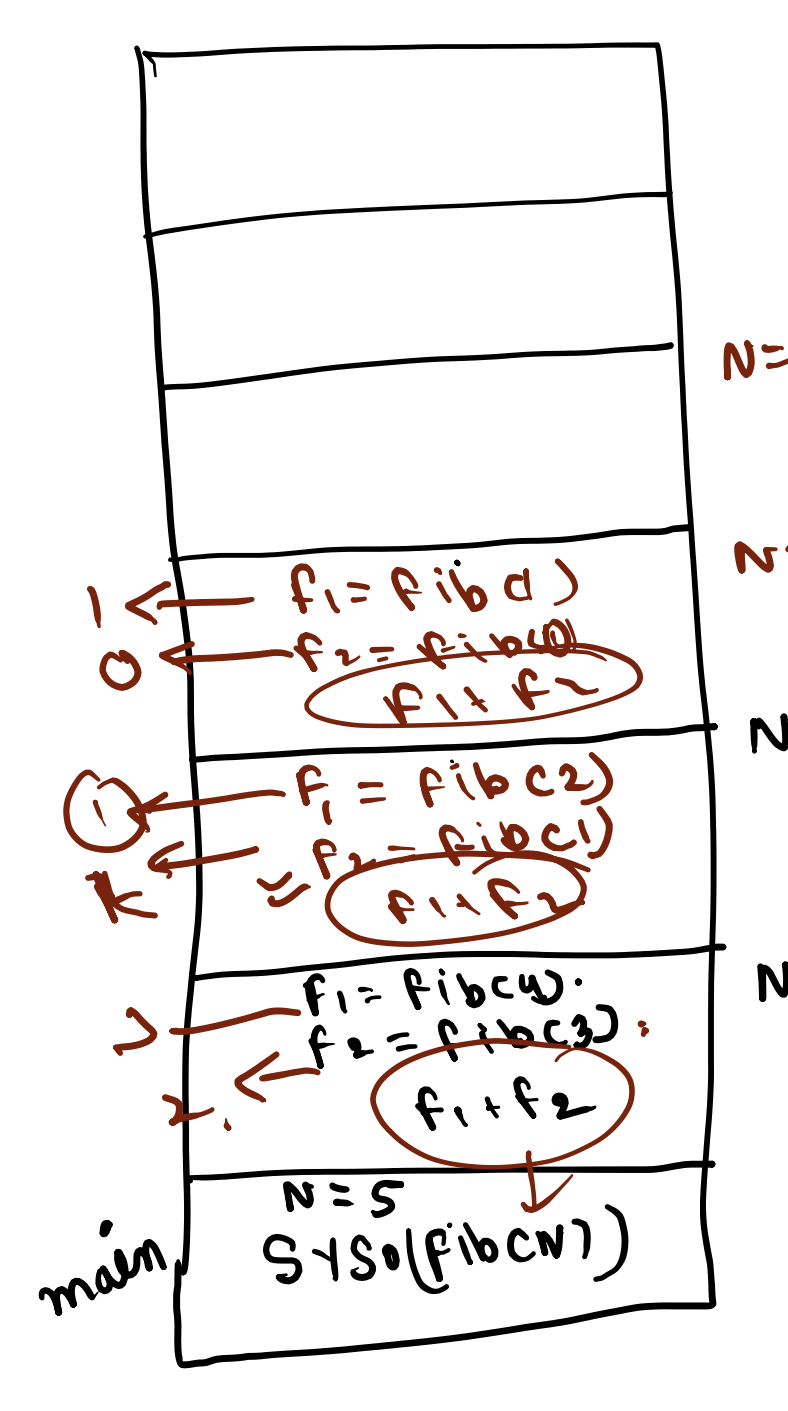
Recursion

```java
public static void main(String[] args) {
        // TODO Auto-generated method stub
        int n = 5;
        System.out.println(fib(n));

    }

    public static int fib(int n) {
        if (n == 0 || n == 1) {
            return n;
        }
        int f1 = fib(n - 1);
        int f2 = fib(n - 2);
        return f1 + f2;
    }
```





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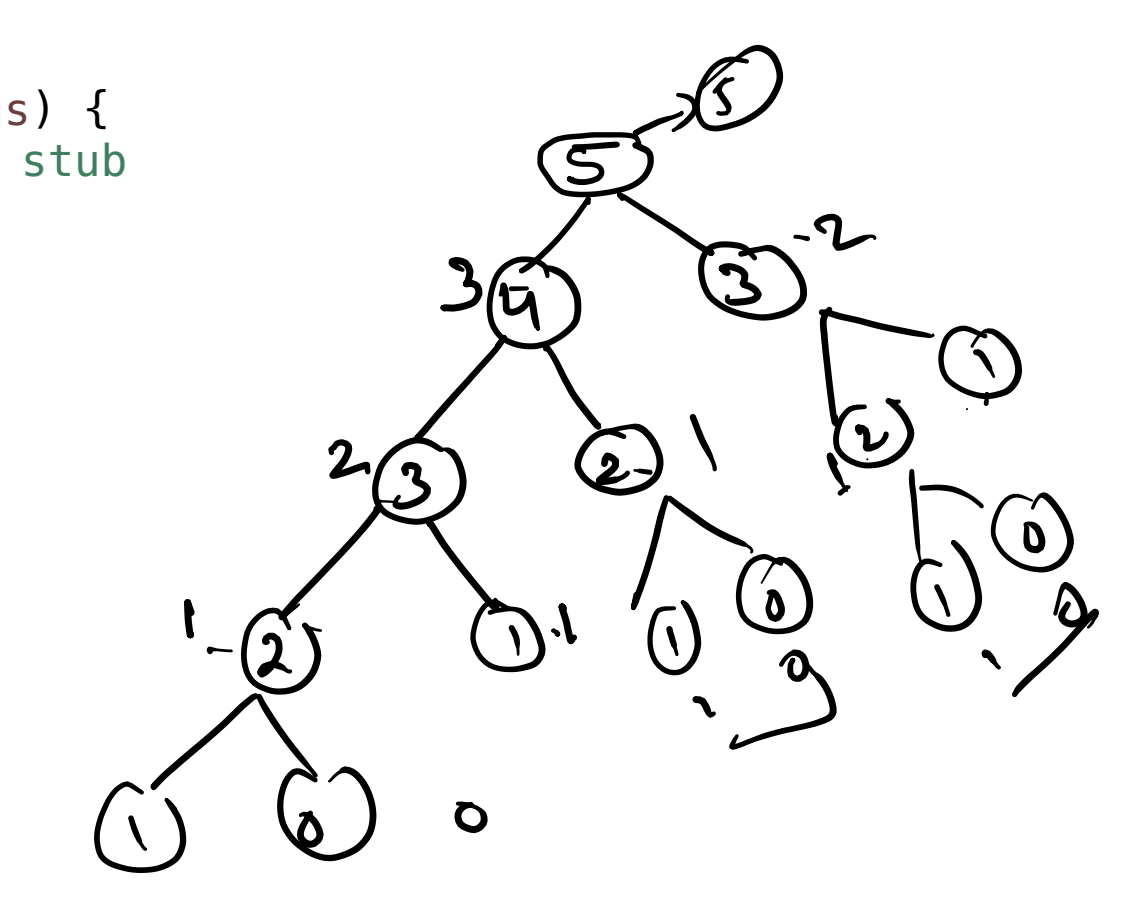

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①

abcd → bcd

bd → x

N

2

abc → { a bc c b ac abc ab }

a [ a ]

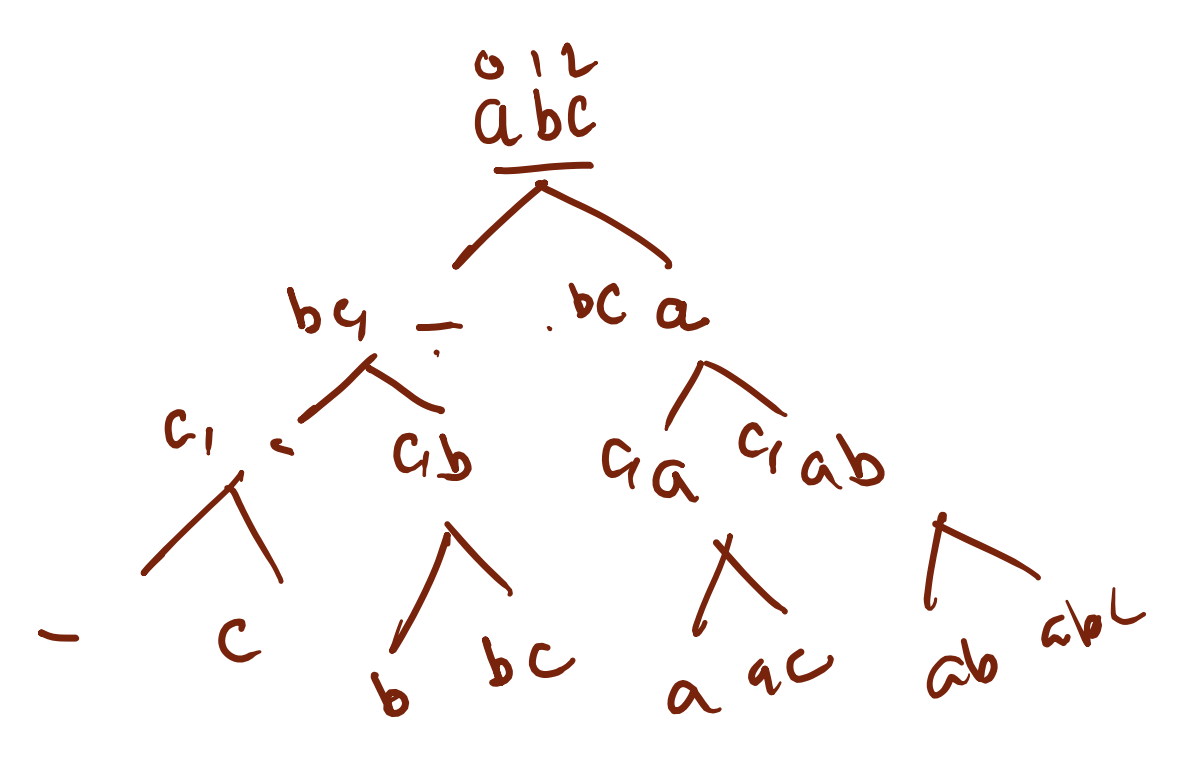
ab = { b a ab }

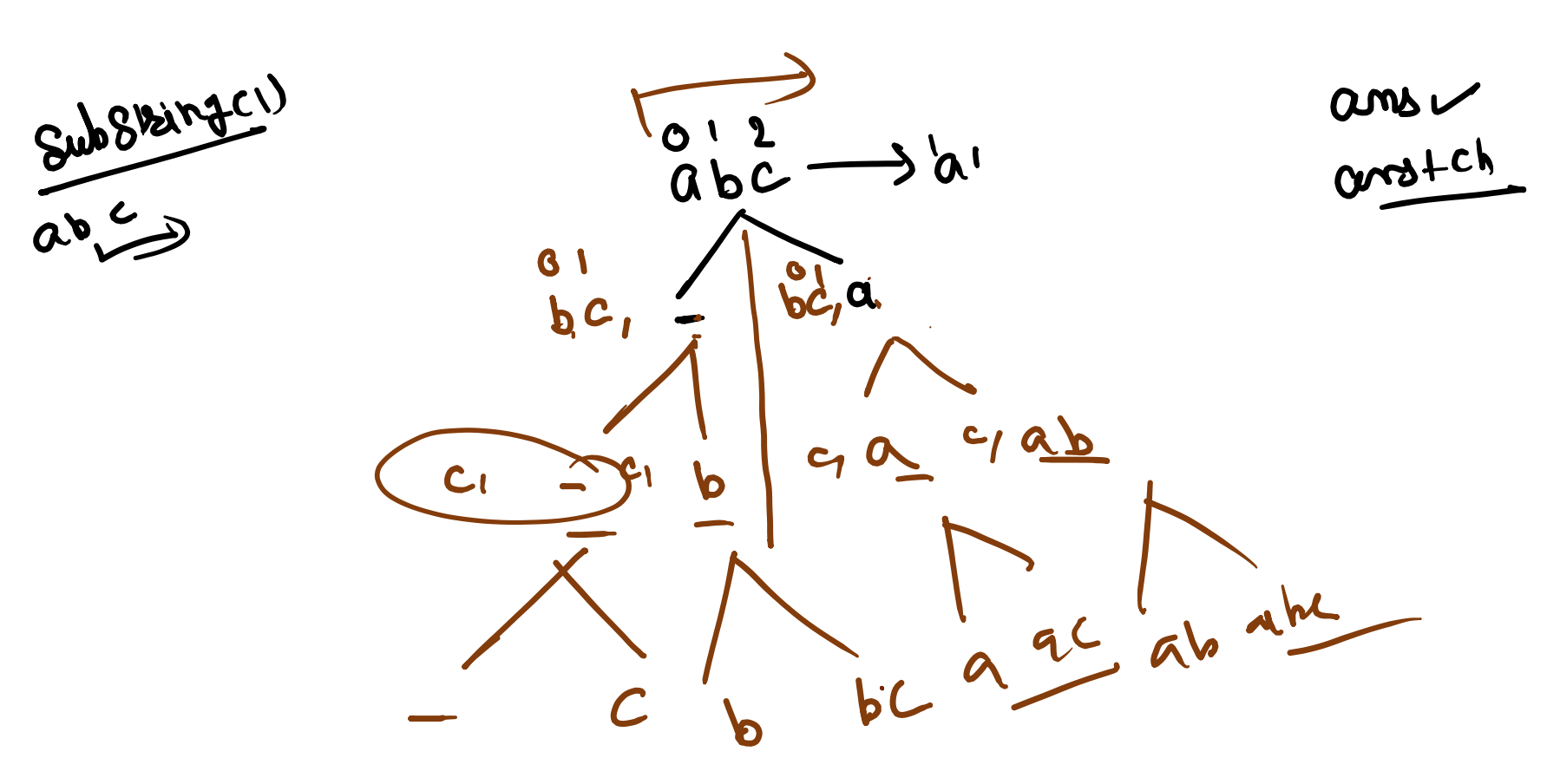
abc = { a c ac b bc ab abc }

— + b = b

a + b = ab

{a} {a b ab}



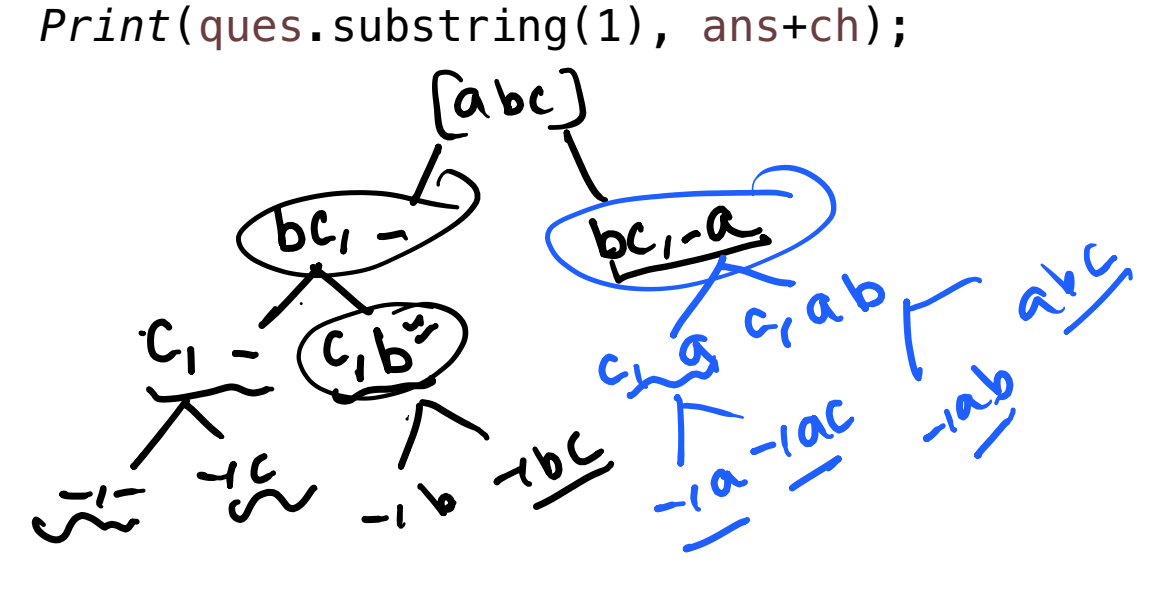


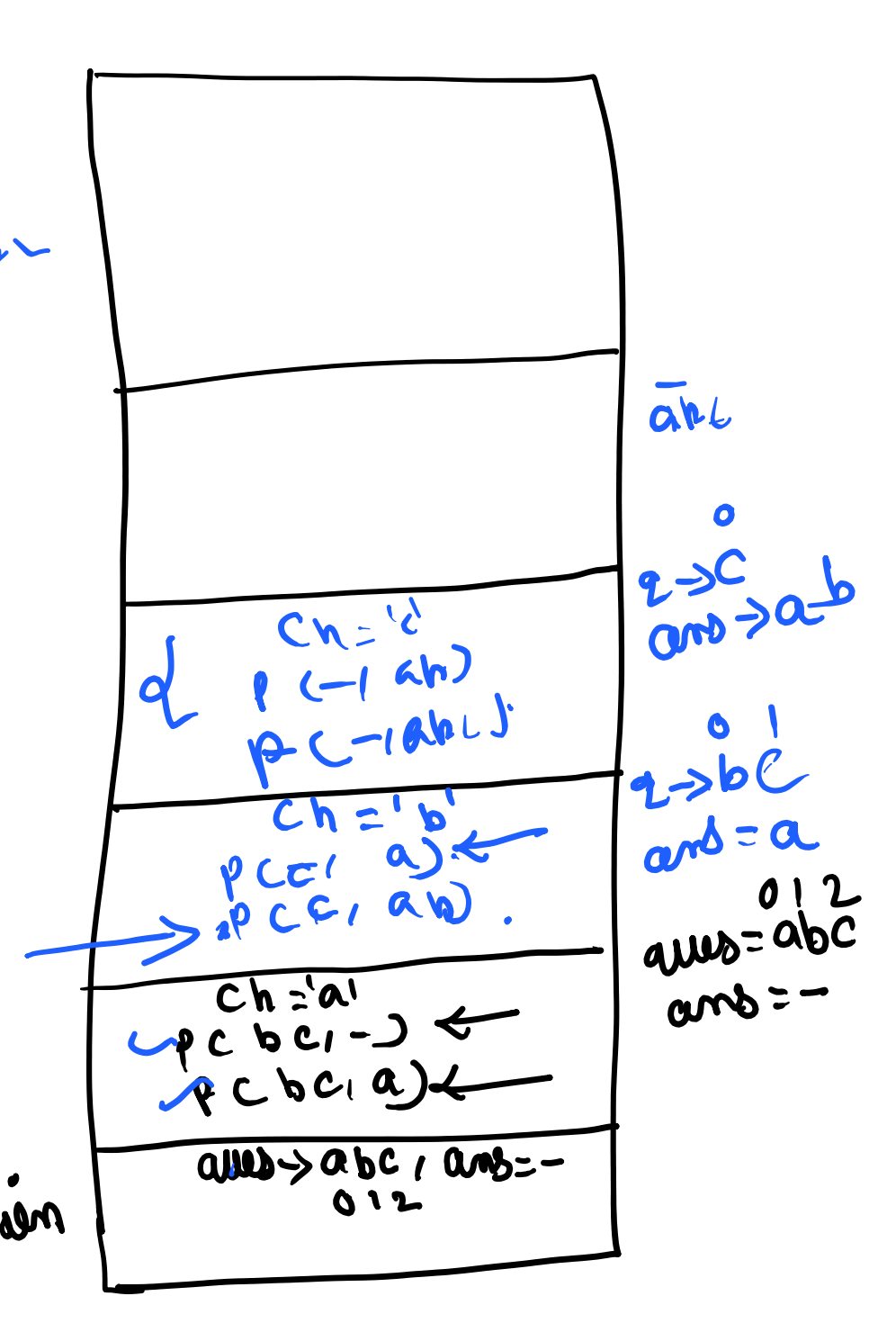
```java
public static void main(String[] args) {
        // TODO Auto-generated method stub
        String ques = "abc";
        Print(ques,"");
    }

    public static void Print(String ques, String ans) {
        // TODO Auto-generated method stub
        if(ques.length()==0) {
            System.out.println(ans);
            return;
        }
        char  ch =ques.charAt(0);
        Print(ques.substring(1), ans);
        Print(ques.substring(1), ans+ch);

    }
```

— , c , b , bc , a , ac , ab , abc





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```

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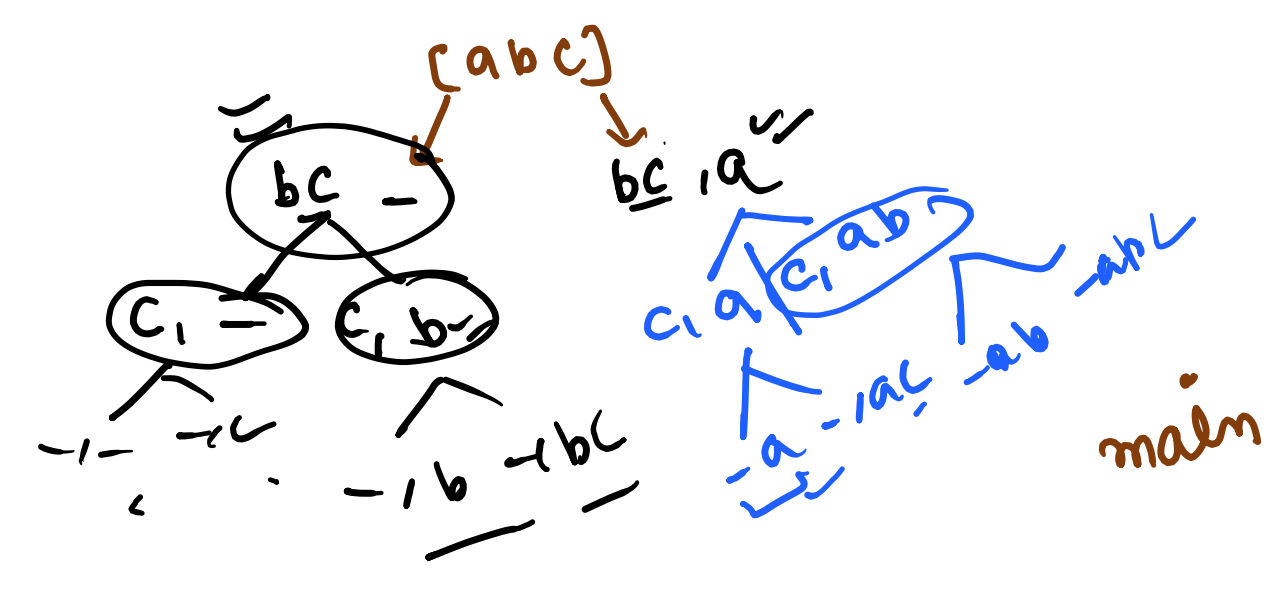

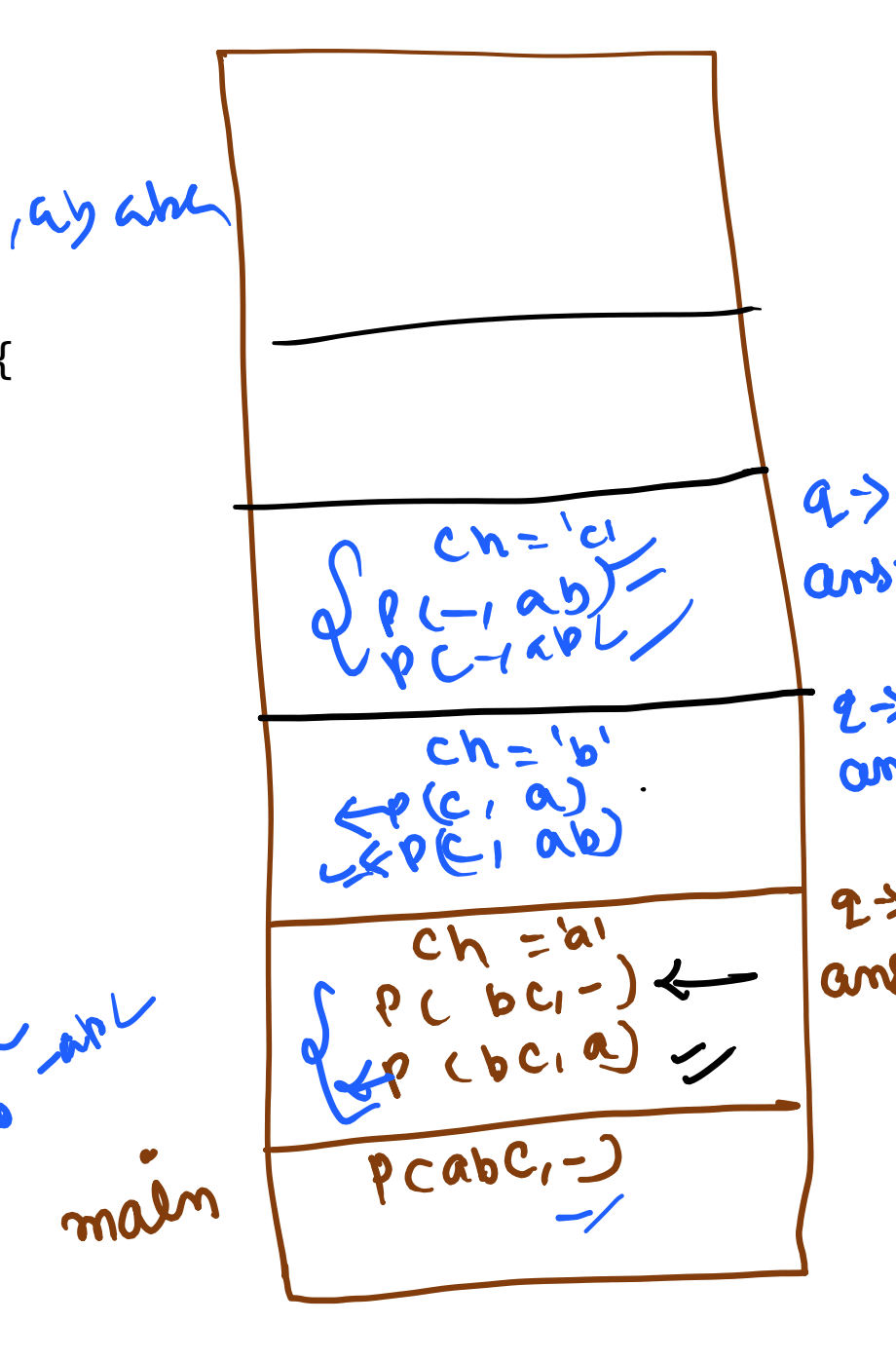

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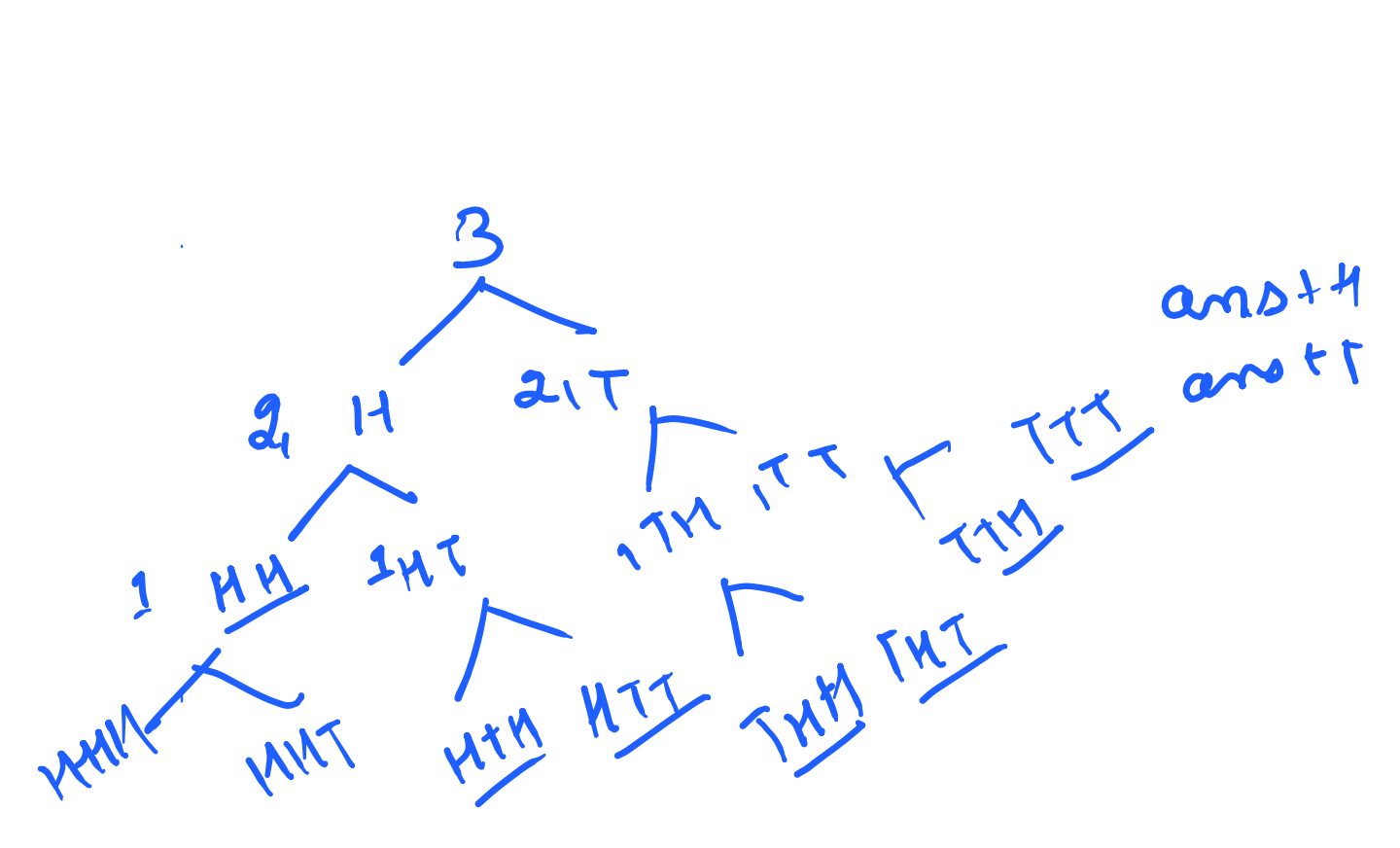

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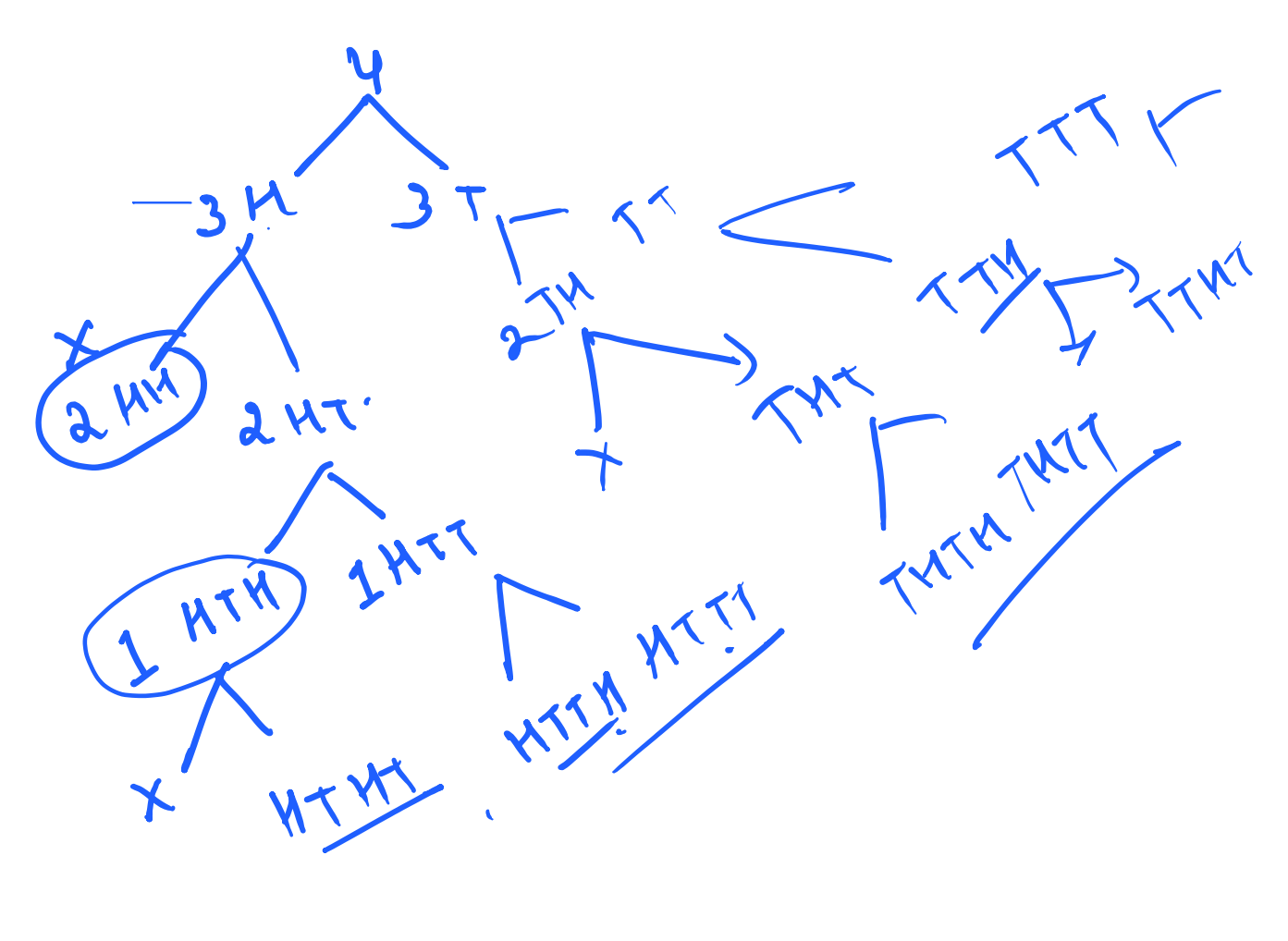
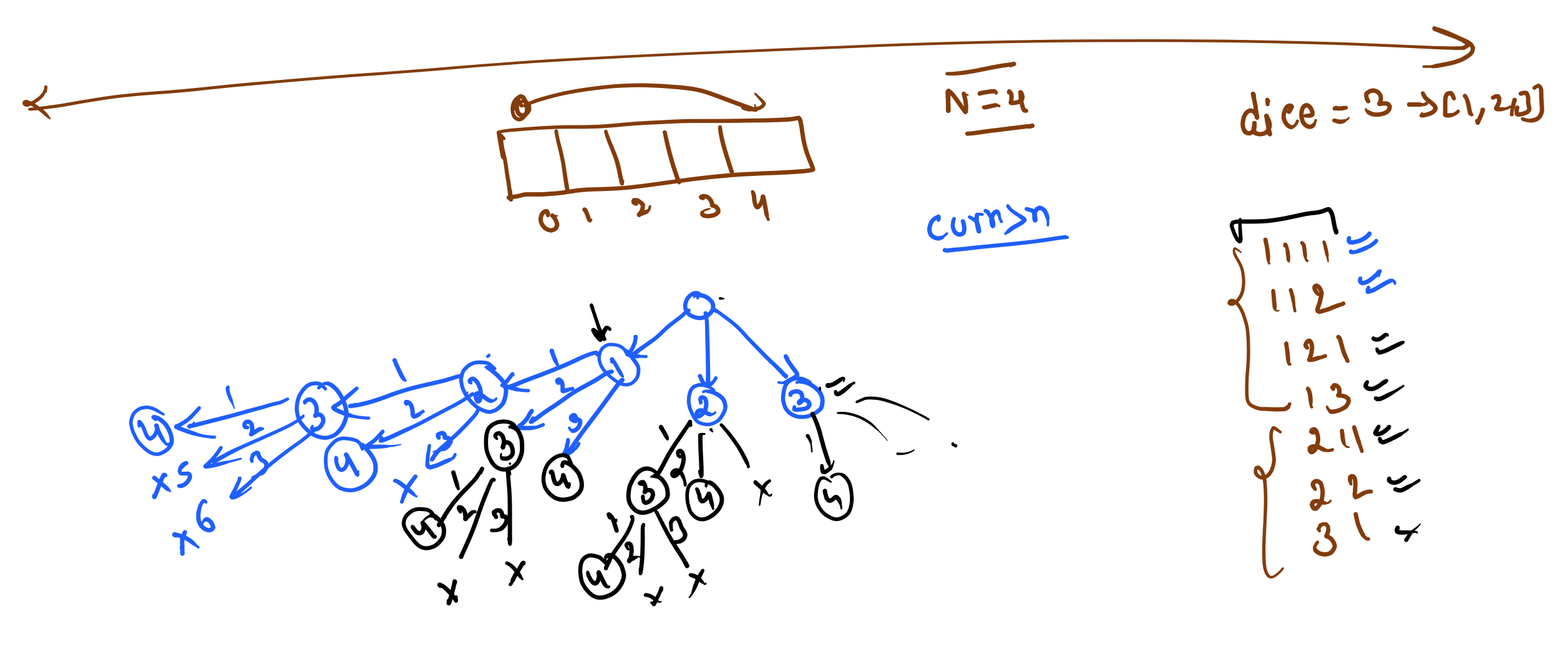


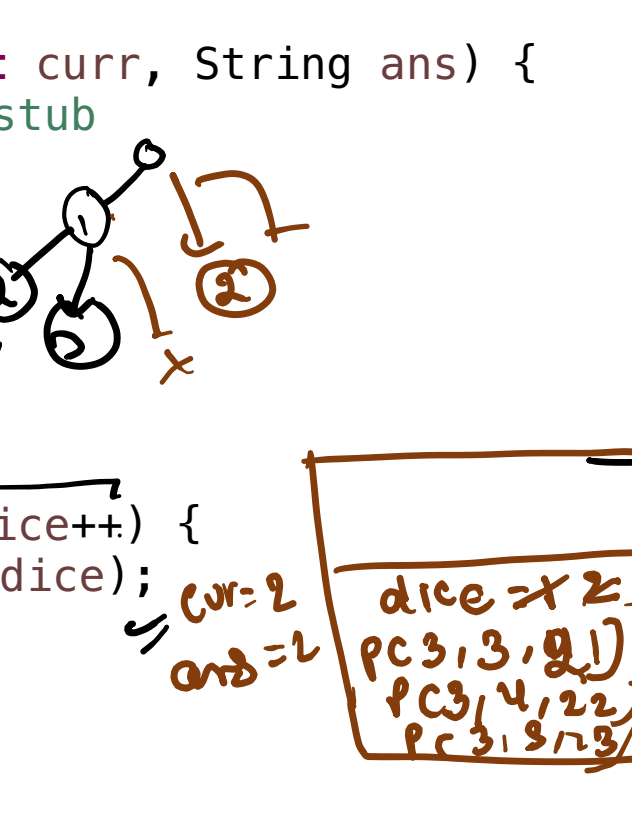


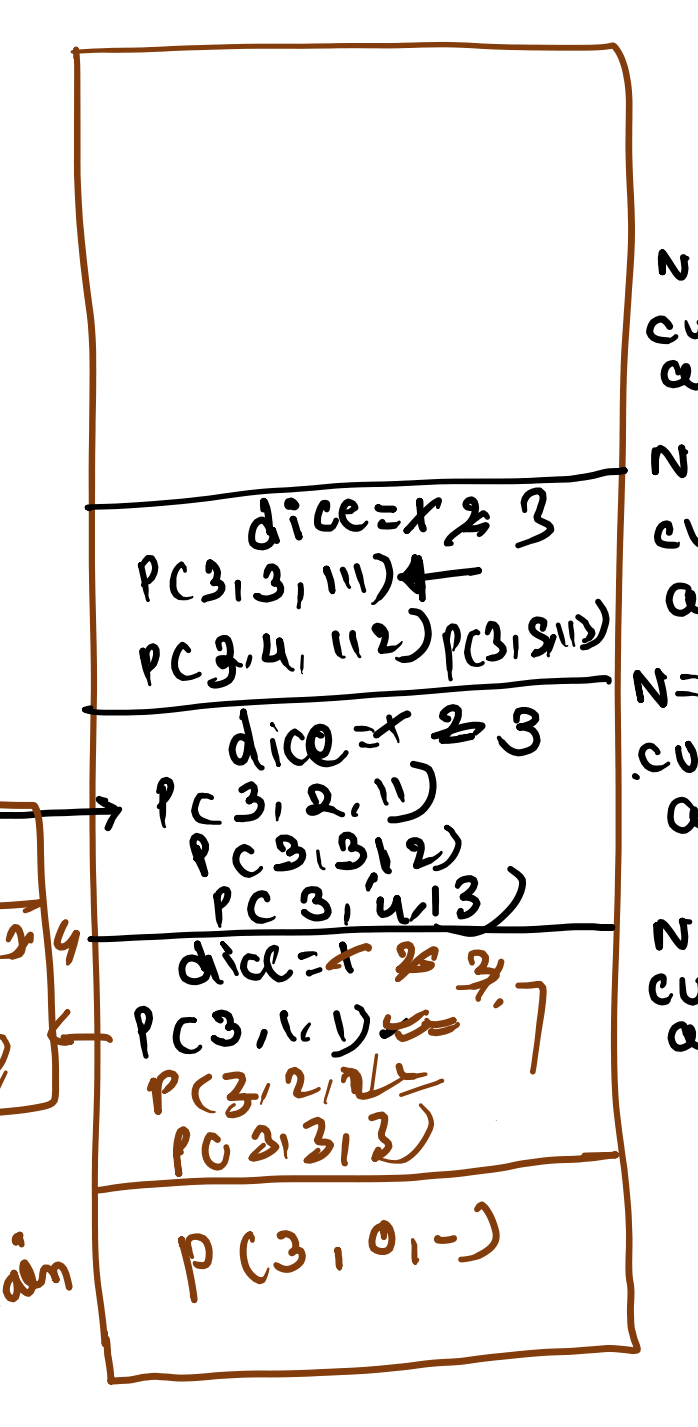
```java
public static void main(String[] args) {
        // TODO Auto-generated method stub
        int n = 3;
        Path(n, 0, "");

    }

    public static void Path(int n, int curr, String ans) {
        // TODO Auto-generated method stub
        if (curr == n) {
            System.out.println(ans);
            return;
        }
        if (curr > n) {
            return;
        }
        for (int dice = 1; dice <=3; dice++) {
            Path(n, curr + dice, ans + dice);
        }
    }
```

111

12

21





```java
public static int Print(String ques, String ans) {
        // TODO Auto-generated method stub
        if (ques.length() == 0) {
            System.out.print(ans + " ");
            return 1;
        }
        char ch = ques.charAt(0);
        int x = Print(ques.substring(1), ans);
        int y = Print(ques.substring(1), ans + ch);
        return x + y;

    }
```

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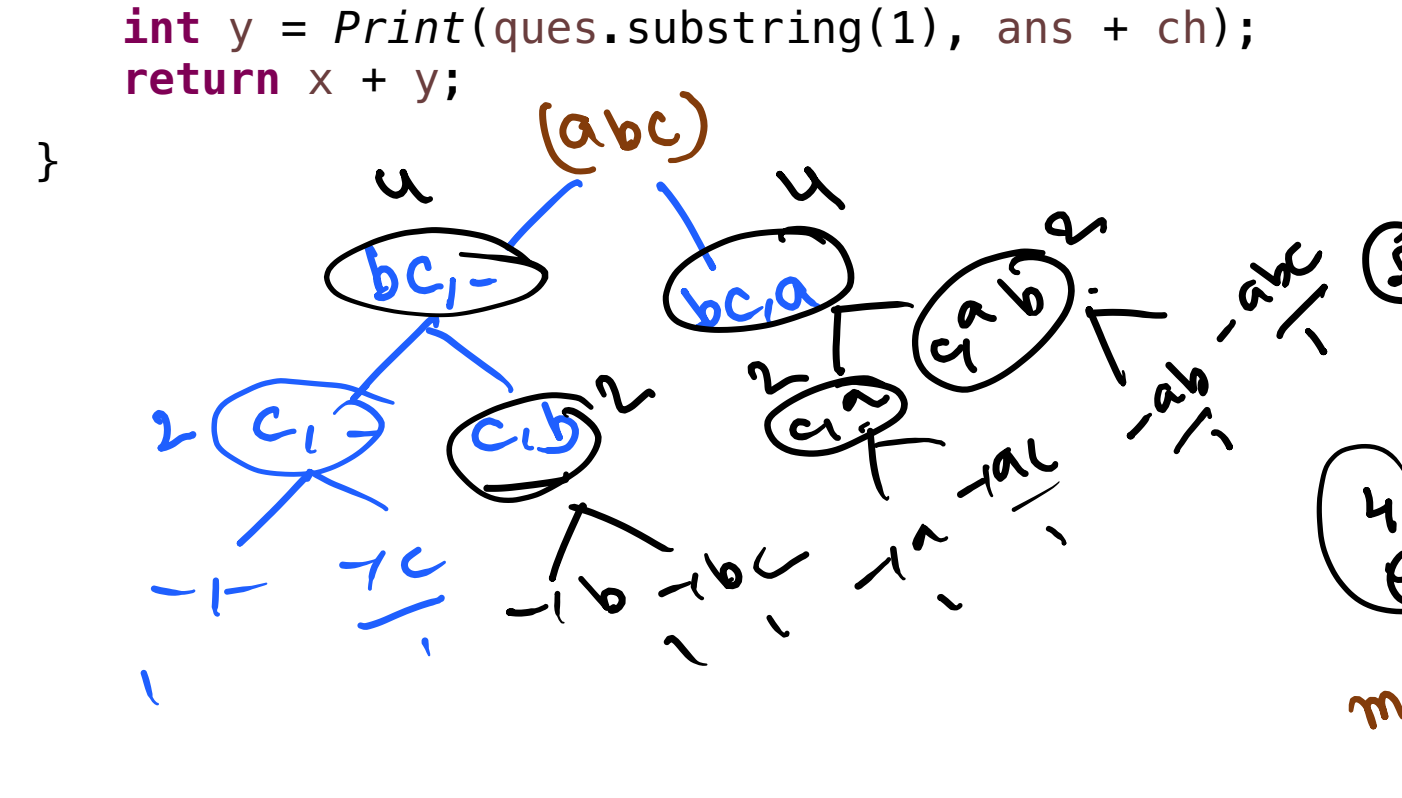

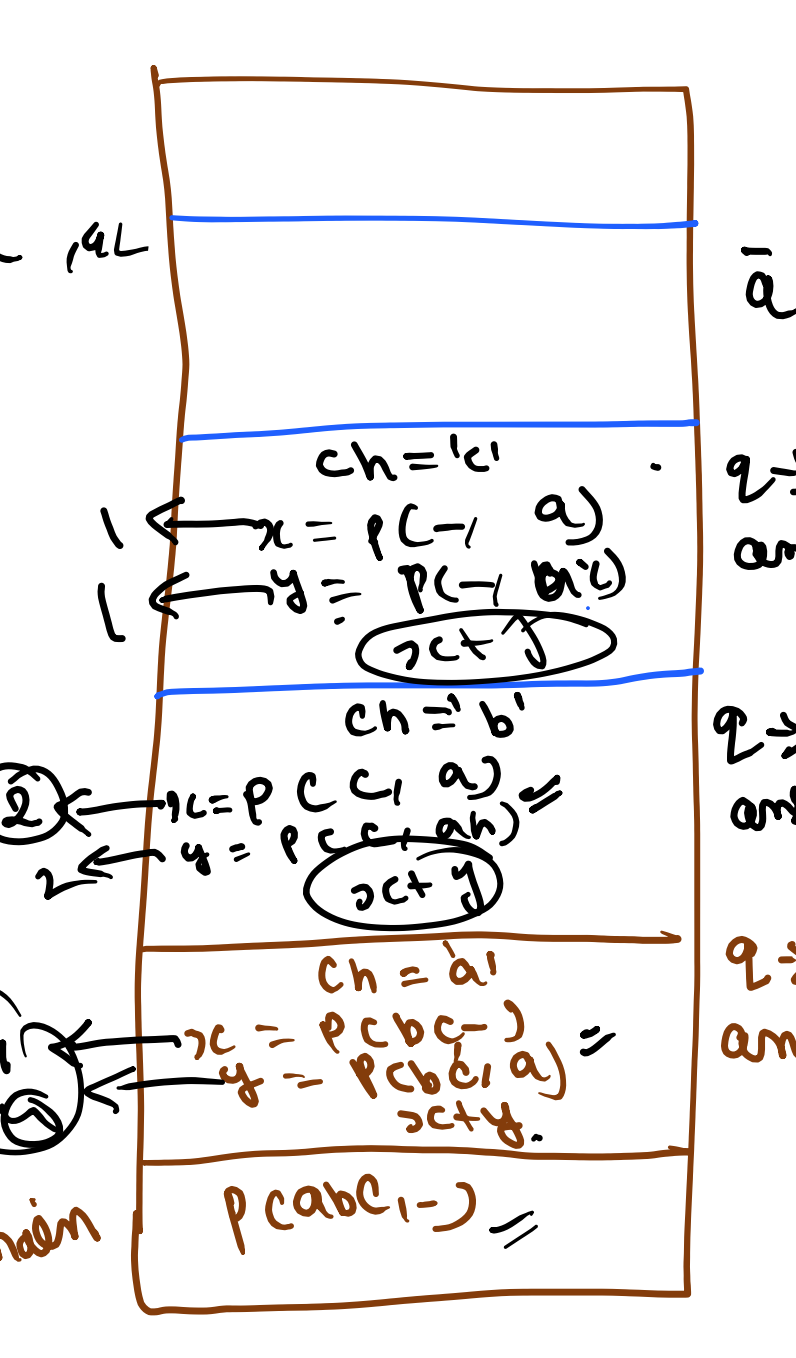

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N=3

((( )))

if(open == n && close == n) {
SYSO ans
}

((()))

(()())

if(open > n || close > open)
return

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()(())
()()()